

AIML ZG528 - Assignment 1: Probabilistic Motion & Sensor Models

Part A - Written Report (Questions 1 & 3)

Domain: Hospitality Bots (Restaurant Food Delivery)
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Question 1: Robot System Description

a) Type of Robot, Function, and Working Environment

Robot Type: Autonomous Mobile Robot (AMR) - a compact wheeled differential-drive delivery robot designed for indoor restaurant environments.

Function: The robot autonomously delivers food plates and beverages from the kitchen counter to dining tables. Its primary tasks include: - Picking up prepared food orders from the kitchen pass/counter. - Navigating between dining tables to reach the correct table number. - Avoiding dynamic obstacles (waiters, guests, chairs pulled out) during delivery.

Working Environment: - **Indoor structured environment:** A rectangular restaurant floor (approx. 10m x 8m) with a kitchen area, dining tables, and walkways between them. - **Semi-dynamic obstacles:** Waiters moving around, guests standing/sitting, chairs partially blocking paths, spilled liquids on floor. - **Mixed floor surfaces:** Rubber mats near the kitchen (good grip), wooden/tile flooring in the dining area (moderate grip), potential wet/spill areas (poor grip). - **Operational constraints:** Must not collide with guests or furniture, must deliver without spilling food, and must operate quietly during service hours.

b) Sensor Selection and Justification

Two Proprioceptive Sensors (internal state measurement):

#	Sensor	Justification
1	Wheel Encoders (Optical Rotary Encoders)	Measure wheel rotations to estimate distance traveled and heading via odometry. Essential for dead-reckoning in a restaurant where GPS is unavailable. Provides the velocity inputs (v , ω) used in our motion model.
2	IMU (Inertial Measurement Unit - Gyroscope + Accelerometer)	Provides angular velocity and linear acceleration data to correct drift in odometry estimates, especially during turns between tables and when wheels slip on wet kitchen floors. Compensates for the rotational drift modeled by α_5 and α_6 parameters.

Two Exteroceptive Sensors (external environment measurement):

#	Sensor	Justification
1	2D LiDAR (e.g., RPLIDAR A2, 360° scan)	Provides range measurements for obstacle detection and localization. Critical for detecting table legs, chairs, walls, and moving people. This is the sensor modeled by our beam-based sensor model (phit, pshort, pmax, prand components). Enables table number recognition, guest detection, and close-range obstacle avoidance. Depth sensing helps detect chair legs and objects at ground level that LiDAR might miss due to its fixed scan height.
2	RGB-D Camera (e.g., Intel RealSense D435)	

Question 2: Motion Model Implementation

See attached Jupyter Notebook file: `Part_A_Motion_Model.ipynb`

The notebook implements: - **Velocity-based motion model** with 6 noise parameters (alpha1-alpha6) - **Beam-based sensor model** with 4-component mixture (hit, short, max, random) - **Trajectory visualization** with 100 Monte Carlo simulations and confidence ellipses - **Combined motion + sensing** demonstration - **Floor condition comparison** showing effect of noise on uncertainty

Question 3: Workplace Evaluation

- The velocity-based probabilistic motion model allows the restaurant delivery bot to account for real-world uncertainties such as wheel slippage on wet kitchen floors and drift when turning between tightly spaced dining tables, enabling more robust path execution than a simple deterministic model.
- The beam-based sensor model ensures reliable obstacle detection by combining ideal measurements with noise handling for unexpected obstacles (a waiter stepping into the path, a chair pulled out), random noise, and max-range readings in open areas.
- Together, these models significantly enhance **safety** by reducing collision probability with guests and furniture -- the robot "knows" it might be slightly off-course and plans accordingly.
- From a **productivity** standpoint, the robot can complete approximately 20-30 deliveries per service shift by following probabilistic paths that account for uncertainty rather than needing constant manual intervention.
- In our workplace context at a restaurant/hospitality business, deploying such a system reduces wait staff workload for routine food runs by ~40%, allowing servers to focus on guest interactions while the robot handles repetitive kitchen-to-table deliveries reliably.